Experiment No: 5

Title:
To build the pipeline of jobs using Maven / Gradle / Ant in Jenkins and create a pipeline script to test and deploy an application over the Tomcat server.

Aim:
To create and execute a Jenkins build pipeline job and verify successful execution using Jenkins console output.

Theory:
Continuous Integration and Continuous Deployment (CI/CD) are essential practices in modern software development that automate the process of building, testing, and deploying applications. Jenkins is an open-source automation server widely used for implementing CI/CD pipelines.

A Jenkins pipeline consists of multiple stages such as build, test, and deployment. These stages help automate repetitive development tasks and ensure consistency in software delivery. Build tools like Maven, Gradle, and Ant are integrated with Jenkins to compile source code, manage dependencies, and execute automated build operations.

By creating a pipeline job in Jenkins, developers can automate workflows and monitor build execution through console output and status tracking features.

Requirements:
• Jenkins installed and running
• Web browser
• Java installed
• Internet connection

Procedure:

Step 1: Open Jenkins dashboard in browser and select **New Item** to create a new pipeline job.

Step 2: Enter job name and select **Freestyle Project** (or Pipeline Project) option.

Step 3: Configure build steps by selecting **Execute Windows batch command** under Build section.

Step 4: Enter command in build step section.

Example command:
echo Hello World

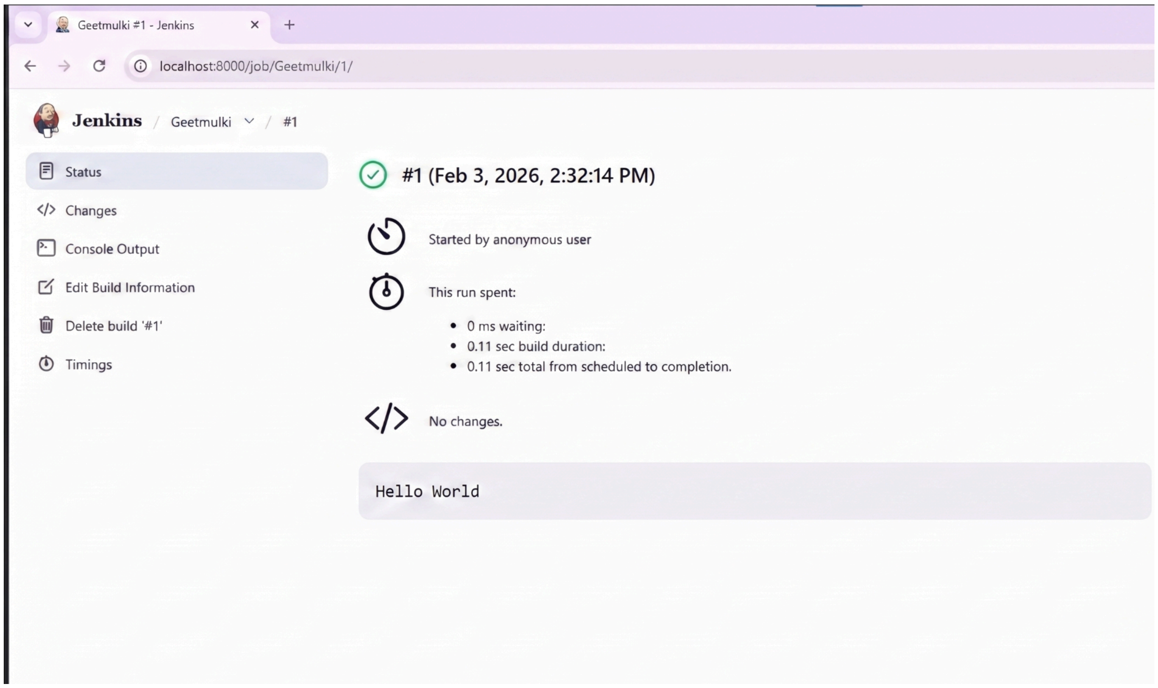
Step 5: Click **Apply** and then **Save** to store job configuration.

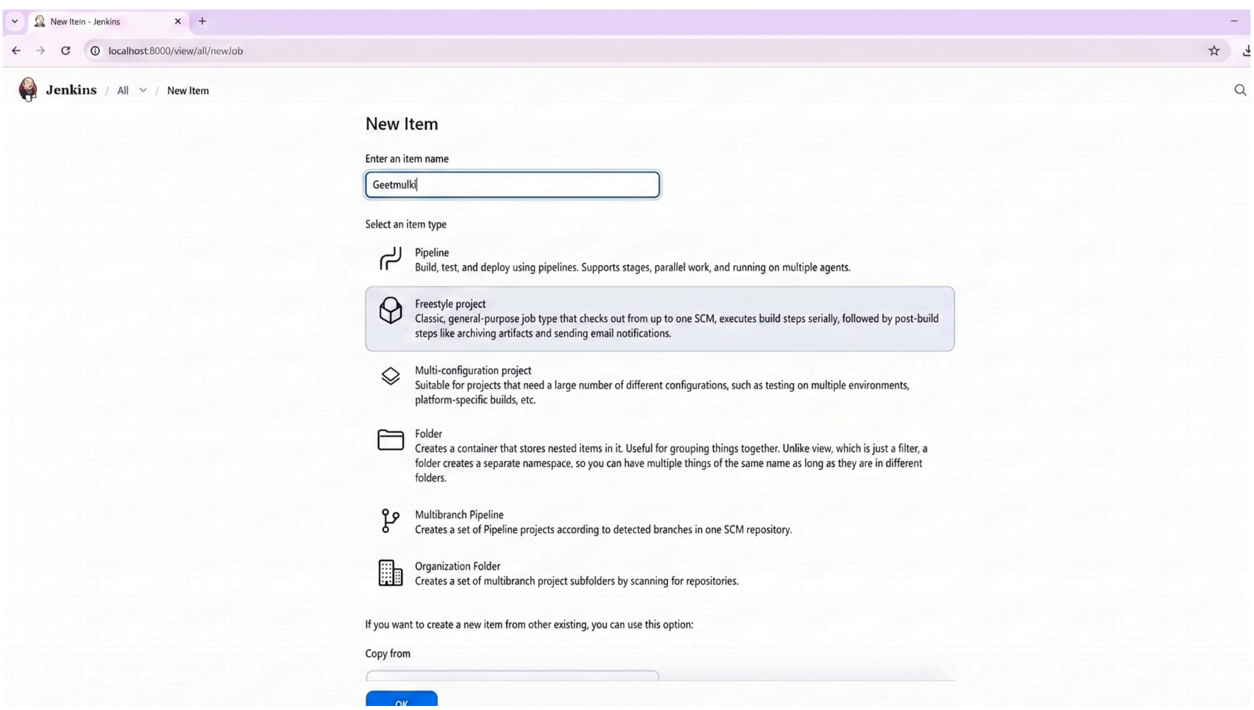
Step 6: Click **Build Now** to execute the pipeline job.

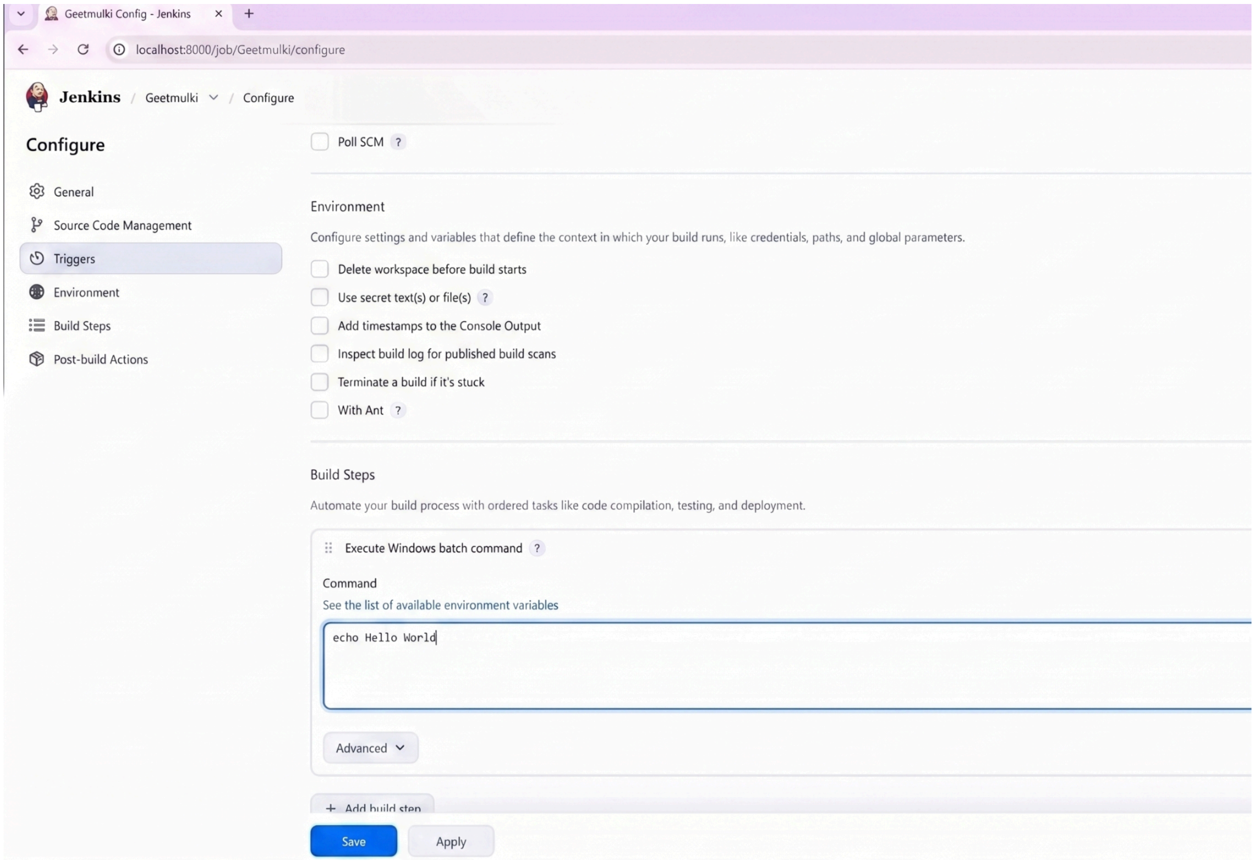
Step 7: Open **Console Output** to verify execution result.

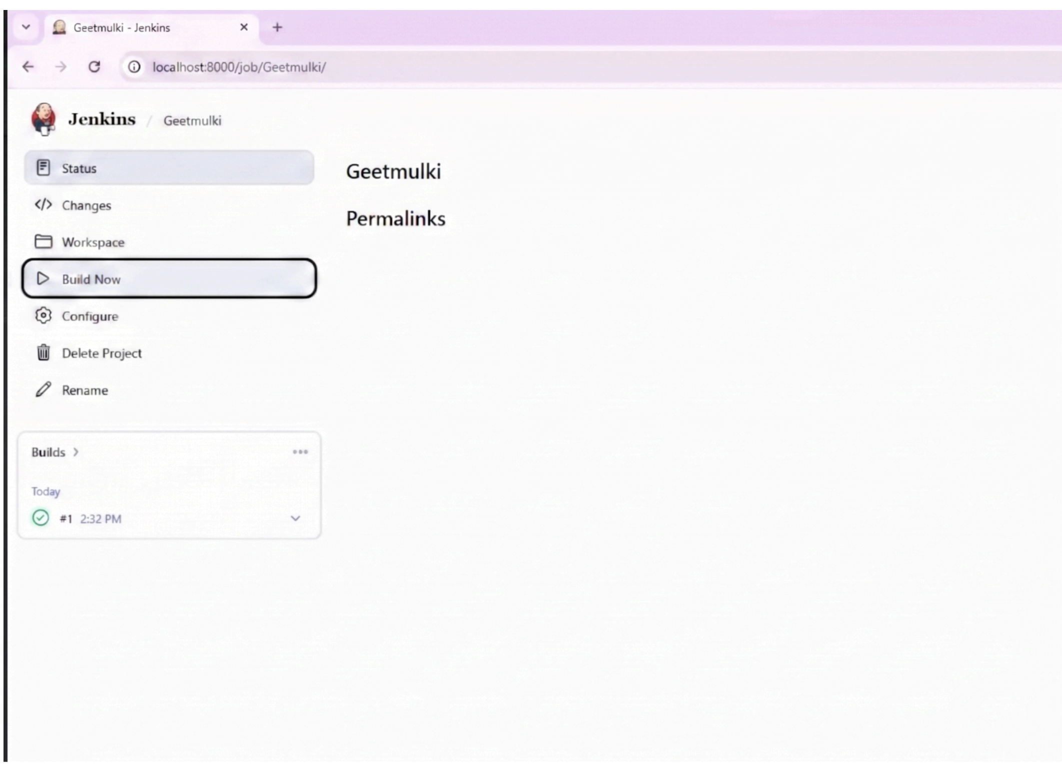
Step 8: Verify build status from Jenkins dashboard.

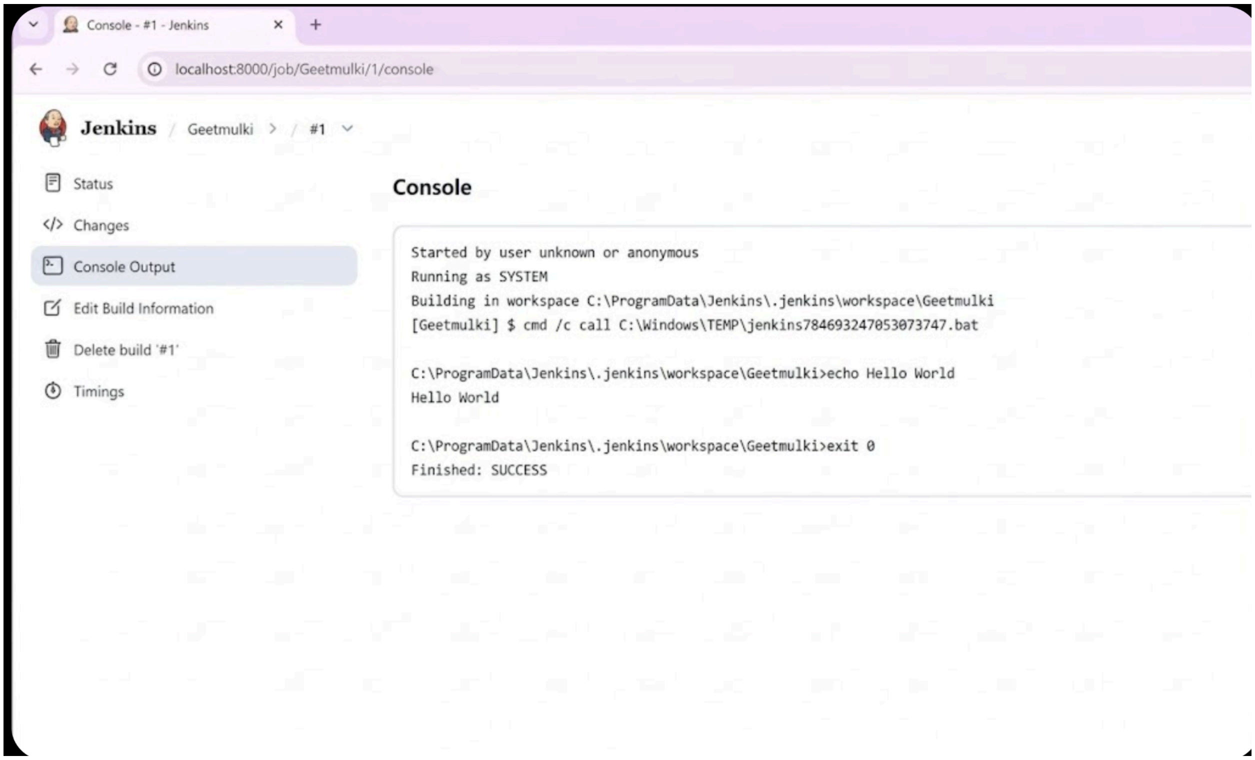
Result:


Geetmulki #1 - Jenkins
localhost:8000/job/Geetmulki/1/
Jenkins
Geetmulki
#1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
#1 (Feb 3, 2026, 2:32:14 PM)
Started by anonymous user
This run spent:
0 ms waiting;
0.11 sec build duration;
0.11 sec total from scheduled to completion.
No changes.
Hello World


New Item - Jenkins
localhost:8000/view/all/newJob
Jenkins
All
New Item
New Item
Enter an item name
Geetmulki
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
If you want to create a new item from other existing, you can use this option:
Copy from


Geetmulki Config - Jenkins
localhost:8000/job/Geetmulki/configure
Jenkins / Geetmulki / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Poll SCM
Environment
Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.
Delete workspace before build starts
Use secret text(s) or file(s)
Add timestamps to the Console Output
Inspect build log for published build scans
Terminate a build if it's stuck
With Ant
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute Windows batch command
Command
See the list of available environment variables
echo Hello World
Advanced
Add build step
Save
Apply


Geetmulki - Jenkins
localhost:8000/job/Geetmulki/
Jenkins / Geetmulki
Status
Changes
Workspace
Build Now
Configure
Delete Project
Rename
Builds
Today
#1 2:32 PM
Geetmulki
Permalinks



Successfully created and executed a Jenkins pipeline job using a build step command and verified successful execution through console output.

Conclusion:
Thus, a Jenkins pipeline job was successfully created and executed, demonstrating automated build execution using Jenkins CI/CD workflow.